

575 Paper Analysis Project

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1. Paper to Analysis

The title of the paper to be analyzed is “Parallel acceleration of CPU and GPU range queries over large data sets” which was accepted by Professor Mike Bailey.

2. General Theme of the Paper

The general theme of the paper is the enhancement of range query performance over large datasets by using parallel processing on CPUs and GPUs. The authors explore the benefits of bitmap indices, specifically utilizing Word-Aligned Hybrid (WAH) compression, for efficient data querying. By using the parallel processing of GPUs, they aim to significantly speed up the execution of these queries.

3. Title Meaning

The title "Parallel acceleration of CPU and GPU range queries over large data sets" indicates that the paper researches the methods for boosting the performance of range queries by parallel execution on both CPUs and GPUs, which is particularly important when dealing with large datasets that would be too time consuming when processed sequentially.

4. Author Information

This paper has contributions from three students at the University of St. Thomas and two professors, one from the University of St. Thomas and one from the University of Puget Sound, all majoring in computer science.

4.1 Mitchell Nelson

Mitchell Nelson was a student at the University of St. Thomas, where he earned the honor of Summa Cum Laude. This paper is his debut in academic publishing. Professionally, Mitchell

worked as an Associate Software Engineer in Minnetonka, Minnesota, and is now employed as a Software Engineer at Canon Medical Informatics.

4.2 Zachary Sorenson

Zachary Sorenson earned his bachelor's degree at the University of St. Thomas and has an impressive professional background as a Software Engineer. He has held roles at NextEra Analytics, served as an HPC (High-Performance Computing) Applications Engineer, and currently works as a Senior Backend Engineer at Custodia Bank.

4.3 Joseph M. Myre

Joseph M. Myre has conducted research at the University of Arkansas, where he used LIDAR, photogrammetry, and Structure from Motion (SfM) to acquire real-world 3D scans of cave systems and developed high-performance lattice-Boltzmann fluid dynamics methods for distributed GPUs. At the University of St. Thomas, Joseph teaches Computer Architecture and C programming and has mentored eight students in HPC research projects. Since October 2023, he has been working as a Staff HPC Applications Engineer at NextSilicon.

4.4 Jason Sawin

Jason Sawin is the Chair and Professor of Computer and Information Sciences at the University of St. Thomas. With over 15 years of academic experience, he has taught more than 85 undergraduate courses and published over 25 peer-reviewed articles, often collaborating with undergraduate students. He earned his Ph.D. in Computer Science from The Ohio State University and has taught students at both the University of Puget Sound and the University of St. His research focuses on improving the efficient processing of large read-only scientific data through advancements in database bitmap indices.

4.5 David Chiu

David Chiu, a professor at the University of Puget Sound, holds both bachelor's and master's degrees in computer science from Kent State University and a Ph.D. from The Ohio State University. His research interests are in data management and cloud computing, particularly in

developing efficient database indices to handle large data sets. David's work is aimed at uncovering valuable information from vast amounts of data.

5. Experiment

To evaluate the performance of GPU and CPU range-query, authors have done a series of tests on Ubuntu 16.04.5, equipped with dual 8-core Intel Xeon E5-2609 v4 CPUs and 322 GB of RAM. The CPU tests were implemented in C++ and compiled with GCC v5.4.0, while the GPU tests were developed using CUDA v9.0.176 and executed on an NVIDIA GeForce GTX 1080 with 8 GB of memory. Authors used six data sets for evaluation, categorized into three real and three synthetic sets. The real data sets included the KDD data set, which is network flow traffic from the KDD Cup'99 containing 4,898,431 rows and 42 attributes, discretized into 475 bins using Lloyd's Algorithm and the Linkage data set, consisting of anonymized epidemiological data from North Rhine-Westphalia with 5,749,132 rows and 12 attributes, discretized into 130 bins; and the BPA data set, which includes power grid measurements from 20 synchrophasors, containing 7,273,800 rows, discretized into 1367 bins. The synthetic data sets were generated using a Zipf distribution with skew values of 0, 1, and 2, each containing 32 million rows and 100 bins.

Both GPU and CPU-based range query were used. The GPU methods included Column-Oriented Access (COA), Row-Oriented Access (ROA), Hybrid, and Ideal Hybrid approaches, while the CPU methods used iterative processing and reduction techniques. For each data set, authors had executed a range query on 64 random bit vectors, ensuring consistency by maintaining a fixed query size. Additionally, they had tested the effect of varying query sizes (4, 8, 16, 32, and 64) using the highest-performing CPU and GPU methods. In the GPU tests, they used the maximum threads per block (32 for ROA, 1024 for others) and the required number of threads blocks per grid. The CPU tests were done with varying core counts (1, 2, 4, 8, and 16 cores). Each experiment was repeated six times, recording the execution time for each trial, while the first result ignored to eliminate the temporal program behavior.

The experiments demonstrated that GPU methods were consistently superior to the CPU method, outperforming it in 96.2% of the tests. The GPU methods achieved an average speedup

of 14.50 $\times$, with a maximum speedup of 54.14 $\times$ for certain data sets. The iterative CPU method showed improvement with the addition of more cores, except for the BPA data set, where performance did not improve when transitioning from 8 to 16 cores. In the comparison between GPU methods and the CPU reduction method, the GPU methods provided average speedups of 7.03 $\times$, 9.47 $\times$, 8.60 $\times$, 7.57 $\times$, 9.46 $\times$, and 9.11 $\times$ for the KDD, linkage, BPA, Zipf (skew = 0), Zipf (skew = 1), and Zipf (skew = 2) data sets. Overall, the GPU methods outperformed the CPU reduction method in all tests, with an average speedup of 8.50 $\times$ and a maximum speedup of 26.01 $\times$. Performance relative to CPU core utilization, the CPU reduction method provided an average speedup of 2.18 $\times$ over the iterative CPU approach when using 16 cores. The reduction method consistently improved performance with most data sets, except for the KDD data set, which experienced a 0.44 $\times$ slowdown due to its highly compressed nature.

The paper evaluated four GPU range query execution strategies which are Column-Oriented Access (COA), Row-Oriented Access (ROA), Hybrid, and Ideal Hybrid. The COA and hybrid methods consistently outperformed the ROA method, with the hybrid methods showing the most significant performance improvements. On average, the COA, hybrid, and ideal hybrid methods provided 25.09%, 38.47%, and 45.23% speedups over the ROA method. The study also showed the impact of varying query sizes on the performance of the methods. It was found that GPU execution times remained relatively consistent compared to CPU times, which grew at a faster rate with increasing query sizes. The ideal hybrid GPU approach demonstrated consistent performance across different query sizes, maintaining a significant speedup over both CPU methods.

6. Insight gained.

It was interesting to see that compressibility affects GPU performance. The paper showed that the relative performance of GPU methods was influenced by the compressibility of the data sets. For the highly compressible KDD data set, CPU methods, performed relatively fine due to effective caching and branch prediction capabilities. However, GPU methods generally excelled

when handling less compressible data, where the CPU's performance was hindered by higher data volume and reduced caching efficiency.

7. Conclusion drawn from this paper.

The experiments demonstrated that GPU-based methods offer significant performance improvements for range query processing over traditional CPU-based methods. The GPU methods, particularly the ideal hybrid approach, showed superior performance by using the highly parallel nature of GPUs and their architectural features, such as coalesced memory accesses and low-latency shared memory.

The study also highlighted the impact of data characteristics on performance, with less compressible data sets were better on GPU methods. The ideal hybrid GPU method consistently provided the best performance, achieving significant speedups over both the iterative and reduction-based CPU approaches. Overall, the findings suggest that GPUs can significantly enhance bitmap indexing and range query processing, making them highly suitable for large-scale and diverse data set applications.

8. Future research

To continue advancing this research, several key steps can be taken. Firstly, exploring the performance of the proposed algorithms on non-NVIDIA GPUs will be essential. Investigating GPUs from other manufacturers will ensure the generalizability of the findings, making the algorithms more versatile across different hardware platforms. Moreover, extending the scalability of these algorithms is crucial. Testing and optimizing the algorithms for even larger data sets and more complex query patterns will help to evaluate their limits and effectiveness in handling big data challenges. Finally, integrating these algorithms with various DBMS will provide insights to build practical system. Assessing how well the algorithms work within different DBMS environments will help to identify potentials of this research, enabling implementation in real-world applications.